

A counterexample to the entropy polynomial root conjecture

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Abstract

Wakhare's entropy polynomial root conjecture asserts that an explicit polynomial $p_{k,r}$ has exactly two roots in $(0, 1)$ for admissible coprime integers $k > r \geq 1$. We give a counterexample with the coprime pair $(k, r) = (11, 10)$. Five exact rational sign evaluations yield at least four distinct roots.

1 The conjecture and the counterexample

In Conjecture 2 of [1], the entropy polynomials are defined by

$$h_{k,r}(x) = \sum_{j=0}^{k-1} x^{rj} \sum_{v=0}^j \frac{(-1)^{j-v}}{v+1} \binom{rv+k}{k} \binom{k}{j-v}, \quad (1)$$

and

$$p_{k,r}(x) = \alpha_{k/r} k (1 - x^r)^k h_{k,k}(x) - r (1 - x^k)^k h_{k,r}(x), \quad (2)$$

where $\alpha_s > 0$ is the unique solution of $\alpha_s(1 + \alpha_s)^{s-1} = 1$. The conjecture states that $p_{k,r}$ has exactly two real roots in $(0, 1)$, counting multiplicity, for the admissible coprime pairs $k > r \geq 1$.

Theorem 1. *The polynomial $p_{11,10}$ has a root in each of*

$$(1/5, 2/5), \quad (2/5, 3/5), \quad (3/5, 2/3), \quad (2/3, 4/5).$$

In particular, this admissible coprime pair refutes Conjecture 2 of [1].

Write $\alpha = \alpha_{11/10}$, so $\alpha^{10}(1 + \alpha) = 1$. Define

$$\begin{aligned} P(t) &= 1 + 352705t + 60632419t^2 + 1227099358t^3 \\ &\quad + 6330005947t^4 + 10701243741t^5 + 6330005947t^6 \\ &\quad + 1227099358t^7 + 60632419t^8 + 352705t^9 + t^{10}, \\ Q(t) &= 11 + 1939817t + 289126442t^2 + 5380098482t^3 \\ &\quad + 25959010187t^4 + 41238382790t^5 + 22851341183t^6 \\ &\quad + 4098130058t^7 + 181139618t^8 + 831413t^9 - t^{10}. \end{aligned}$$

Evaluation of the finite sums in (1) gives exactly

$$h_{11,11}(x) = P(x^{11}), \quad h_{11,10}(x) = Q(x^{10})/11. \quad (3)$$

Both $P(t)$ and $Q(t)$ are positive for $0 < t < 1$: all coefficients of P are positive, while the constant and final terms of Q sum to $11 - t^{10} > 0$ and its other coefficients are positive.

2 Exact signs and the proof

The function $z^{10}(1+z)$ is strictly increasing on $z > 0$. The integer identities

$$\begin{aligned} 117^{10} \cdot 242 - 125^{11} &= -90074807210933370467 < 0, \\ 937^{10} \cdot 1937 - 1000^{11} &= 10474898608767871728104442364513 > 0 \end{aligned}$$

therefore establish

$$117/125 < \alpha < 937/1000. \quad (4)$$

For $0 < x < 1$, put

$$R(x) = \frac{10(1-x^{11})^{11}Q(x^{10})}{121(1-x^{10})^{11}P(x^{11})}. \quad (5)$$

The denominator is positive, and (2)–(3) give

$$p_{11,10}(x) = 11(1-x^{10})^{11}P(x^{11})(\alpha - R(x)).$$

Thus the sign of $p_{11,10}(x)$ is the sign of $\alpha - R(x)$. The strict rational comparisons below are certified by the positive integers in the appendix.

x	Exact bound on $R(x)$	$\text{sgn } p_{11,10}(x)$
1/5	$R(x) < 919/1000 < 117/125$	+
2/5	$R(x) > 1131/1000 > 937/1000$	−
3/5	$R(x) < 935/1000 < 117/125$	+
2/3	$R(x) > 938/1000 > 937/1000$	−
4/5	$R(x) < 928/1000 < 117/125$	+

Since $p_{11,10}$ is continuous and these five ordered points lie in $(0, 1)$, the intermediate value theorem gives a root in each of the four disjoint intervals in Theorem 1. The roots are distinct, proving the theorem. $\square$

Scope. The argument proves *at least four* roots, and makes no assertion that there are exactly four. It does not refute the associated binary-entropy inequality, Conjecture 1 of [1], proved for all real exponents greater than one by Ho [2]. It also does not resolve Frankl’s union-closed sets conjecture.

Reproduction. The accompanying `check_certificate.py` reconstructs (3) from (1), checks (4), and verifies all five strict comparisons using integer arithmetic only. It uses no floating-point arithmetic, computer algebra system, or numerical root finder. Full numerator and denominator data are included in `certificate.json`.

A Cleared denominator certificate

Write $P(t) = \sum_{j=0}^{10} p_j t^j$ and $Q(t) = \sum_{j=0}^{10} q_j t^j$. For integers $0 < a < b$, define

$$\begin{aligned} \mathcal{P}(a, b) &= \sum_{j=0}^{10} p_j a^{11j} b^{11(10-j)}, & \mathcal{Q}(a, b) &= \sum_{j=0}^{10} q_j a^{10j} b^{10(10-j)}, \\ N(a, b) &= 10(b^{11} - a^{11})^{11} \mathcal{Q}(a, b), & D(a, b) &= 121b(b^{10} - a^{10})^{11} \mathcal{P}(a, b). \end{aligned}$$

Then $N, D > 0$ and $R(a/b) = N(a, b)/D(a, b)$. The following are exact positive integer values. Each decimal integer is wrapped across lines; its lines are to be concatenated.

$$919D(1, 5) - 1000N(1, 5) =$$

46687482475003686731728494443175716091029147392029835848827061539678
77835881726361080687327974249063360575405536849747865170141728596492
75824805705028730880

$$1000N(2, 5) - 1131D(2, 5) =$$

42538194776507513458711885524826212893895489706184796164361512069476
99426927690709188274361013225559293120108639340014430569971253115788
9439015076159529241505

$$935D(3, 5) - 1000N(3, 5) =$$

52516379104498148655863274985954973433649161277407686847051045991991
85256340399027956110708767358049927938282779151210248609234143827247
884850356820630516531200

$$1000N(2, 3) - 938D(2, 3) =$$

88008107586949258139134527029175262251555372081468008081118660645776
7956959351946751016842608896901672363281250

$$928D(4, 5) - 1000N(4, 5) =$$

12136820388591534024573035987433025741626348214182265217990319570111
14963599358661651488660016203235762863004128345364061271715446024425
596394854034905941742192560

B Formal verification and reproduction

The principal theorem has a complete Lean proof. The precise scope is: The $k=11$, $r=10$ counterexample with the original real-power parameter, its bridge to the integer equation, exact coefficients, strict signs and four distinct open-interval roots. No claim about the exact root count or the separate entropy inequality. The public proof source snapshot is fixed at commit 02af7db99db789fecb0d5f5f8f37b1a4e40762a0 in <https://github.com/mxym/math>. The proof packages, with their own dependency locks, audit records and reproduction instructions, are:

- `notes/entropy-polynomial-counterexample/formalization`.

The main formal entry points are:

- `EntropyCounterexample.counterexample_to_conjecture_two_original_real_power`.
- `EntropyCounterexample.wakhare_conjecture_two_false_original_real_power`.

Lean is fixed to version 4.34.1, compiler commit 5045d0056413266e57c625dcd7c365b10e377c52; Mathlib is fixed to d13f23b723b8a846827a245b89c10fc7d3f11612. The audited mathematical roots depend only on the standard axioms `prope`, `Classical.choice`, and `Quot.sound`. The archived audits state their actual compilation and replay scopes; publication of this typeset version does not imply a new execution of every previous empty-kernel replay. The companion README maps each printed principal result to its formal theorem and explains how to repeat verification.

An additional independent five-module project is fixed at commit 93a7512b3606581fcbbca310c7d2d587a8c1a903 in `formalizations/wakhare-entropy-four-roots-lean`. Its declaration `EntropyRoot.actual_polynomial_has_four_distinct_roots` constructs an actual real polynomial directly from the original binomial sums and proves four strictly ordered interior roots. The algebraic parameter’s uniqueness and all five strict rational sign certificates are included. The earlier package separately provides the original real-power parameter bridge. The new package’s README and replay script identify the complete build and axiom audit; its recorded clean CI run is <https://github.com/mxym/math/actions/runs/37853632676>. This article revision archives both implementations and does not claim that the later project was present in the earlier version’s DOI.

Research and preparation disclosures

This research, formalization and manuscript preparation used AI assistance. The author is responsible for the manuscript. The principal claims are supported by the readable proofs and the public Lean artifacts specified above. Formal verification is not external human peer review; no such review is claimed. No external funding was received. No worldwide priority or journal acceptance is asserted.

References

- [1] T. Wakhare, *Iterated entropy derivatives and binary entropy inequalities*, Journal of Approximation Theory **307** (2025), 106143. doi:10.1016/j.jat.2025.106143. arXiv:2312.14743v2.
- [2] B. S. Ho, *A generalization of Boppana’s entropy inequality*, arXiv:2601.19327 (2026).